

一、

(1) 光电效应：光照射到金属表面导致大量电子从金属表面逸出的现象

受激辐射：处于激发态的发光原子在外来辐射场的作用下，向低能态或基态跃迁时，辐射光子的现象

(2) Stark 效应：原子或分子在外电场作用下能级和光谱发生分裂的现象

Zeeman 效应：原子在外磁场中发光谱线发生分裂且偏振的现象

(3) 1

(4) $\Delta x \Delta p \geq \frac{\hbar}{2}$ ，不可能同时精确确定一个基本粒子的位置和动量

若两个力学量不对易，则它们不能同时测准

$$(5) E_n^{(1)} = \sum_{m \neq n} \frac{H_{nm}^{(1)}}{E_n^{(0)} - E_m^{(0)}}$$

(6) 量子隧穿效应

$$(7) \hat{L}_x = y\hat{p}_z - z\hat{p}_y \quad \hat{L}_y = z\hat{p}_x - x\hat{p}_z \quad \hat{L}_z = x\hat{p}_y - y\hat{p}_x$$

$$[\hat{L}_x, \hat{L}_y] = [y\hat{p}_z - z\hat{p}_y, z\hat{p}_x - x\hat{p}_z] = [y\hat{p}_z, z\hat{p}_x] + [y\hat{p}_z, -x\hat{p}_z] + [-z\hat{p}_y, z\hat{p}_x] + [-z\hat{p}_y, -x\hat{p}_z]$$

$$= y[z\hat{p}_z, \hat{p}_x] + x[z, \hat{p}_z]\hat{p}_y = -y\hbar\hat{p}_x + x\hbar\hat{p}_y = \hbar(x\hat{p}_y - y\hat{p}_x) = \hbar\hat{L}_z$$

$$\text{同理: } [\hat{L}_y, \hat{L}_z] = \hbar\hat{L}_x \quad [\hat{L}_z, \hat{L}_x] = \hbar\hat{L}_y$$

二、

$$(1) -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} - V_0\psi = E\psi \Rightarrow \psi'' = -\frac{2m(E+V_0)}{\hbar^2} \psi \quad \text{令 } \beta = \sqrt{\frac{2m(E+V_0)}{\hbar^2}} \quad 0 < x < a \text{ 时}$$

$$\therefore \psi_1 = A \sin \beta x + B \cos \beta x \quad \because \psi_1(0) = 0 \Rightarrow B = 0 \Rightarrow \psi_1 = A \sin \beta x$$

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \Rightarrow \psi'' = -\frac{2mE}{\hbar^2} \psi \quad \text{令 } \alpha = \sqrt{\frac{2mE}{\hbar^2}} \quad x > a \text{ 时}$$

$$\therefore \psi_2 = C e^{i\alpha x} + D e^{-i\alpha x} \quad \text{即 } \psi = \begin{cases} 0 & x < 0 \\ A \sin \beta x & 0 < x < a \\ C e^{i\alpha x} + D e^{-i\alpha x} & x > a \end{cases}$$

在 $x=0$ 处， ψ 和 ψ' 连续，得

$$\begin{cases} A \sin \beta a = C \sin \alpha a + D \cos \alpha a \\ \beta A \cos \beta a = \alpha C \cos \alpha a - \alpha D \sin \alpha a \end{cases} = \begin{cases} C e^{i\alpha a} + D e^{-i\alpha a} \\ i\alpha C e^{i\alpha a} - i\alpha D e^{-i\alpha a} \end{cases}$$

$\Rightarrow E > 0$ 时能级所满足的方程

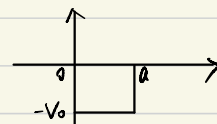
$$\text{若为束缚态, } \psi_2(+\infty) = 0 \Rightarrow C = 0 \Rightarrow \psi_2 = D e^{-\alpha x} \quad (E < 0)$$

$$\begin{cases} A \sin \beta a = D e^{-\alpha a} \\ \beta A \cos \beta a = -\alpha D e^{-\alpha a} \end{cases} \Rightarrow \beta \cot \beta a = -\alpha \Rightarrow \beta a \cot \beta a = -\alpha a$$

$$\text{令 } \beta a = \eta \quad \alpha a = \gamma \Rightarrow \eta \cot \eta = -\gamma$$

0) 束缚态, $E < 0$, $\eta \cot \eta = -\gamma$

$$\eta^2 + \gamma^2 = (\beta^2 + \alpha^2) a^2 = \left(\frac{\pi}{2}\right)^2 \Rightarrow V_0 \geq \frac{\pi^2 \hbar^2}{8ma^2}$$



三、

$$\hat{H} = \frac{\hat{p}_x^2}{2m} - \frac{e^2}{r} - \alpha \frac{e^2}{r} \quad \hat{H}_0 = \frac{\hat{p}_x^2}{2m} - \frac{e^2}{r}, \quad \hat{H}' = -\alpha \frac{e^2}{r} \quad \psi_0 = \sqrt{\frac{1}{\pi a_0^3}} e^{-r/a_0}$$

$$\therefore E_1^{(1)} = \int \psi_0 \hat{H}' \psi_0 d\tau = \int_0^\infty \frac{1}{\pi a_0^3} e^{-2r/a_0} \cdot \left(-\alpha \frac{e^2}{r}\right) \cdot 4\pi r^2 dr = -\frac{2\alpha e^2}{a_0}$$

$$E_1 = E_1^{(0)} + E_1^{(1)} = -\frac{me^4}{2\hbar^2} - \frac{2\alpha e^2}{a_0} = -\frac{e^2}{2a_0} - \frac{2\alpha e^2}{a_0}$$

四、

- (1) 势能相比原来降低, 所以能级减少, 当 $E > V_0$ 时, 出现散射态
 (2) 势能相比原来增加, 所以能级会提高, 当能级很高时, 趋于无限深势阱, $E_n \sim n^2$
 能级随 n 的关系介于一次与二次之间, n 越大, 越趋于二次。

五、

(1) 基态: $\psi_0 = \psi_1(r_1) \psi_2(r_2)$

第一激发态: $\psi_1 = \frac{1}{\sqrt{2}} [\psi_1(r_1) \psi_2(r_2) + \psi_2(r_1) \psi_1(r_2)]$

(2) $\because S=1 \quad \therefore S_z=0, 1, 2$

$\therefore$ 自旋对称: $|0, 0\rangle = \frac{1}{\sqrt{2}} |1, -1\rangle - \frac{1}{\sqrt{2}} |0, 0\rangle + \frac{1}{\sqrt{2}} |1, 1\rangle$

$|2, 2\rangle = |1, 1\rangle \quad |2, -2\rangle = |-1, -1\rangle$

$|2, 1\rangle = \frac{1}{\sqrt{2}} |1, 0\rangle + \frac{1}{\sqrt{2}} |0, 1\rangle$

$|2, -1\rangle = \frac{1}{\sqrt{2}} |-1, 0\rangle + \frac{1}{\sqrt{2}} |0, -1\rangle$

$|2, 0\rangle = \frac{1}{\sqrt{6}} |1, -1\rangle + \frac{1}{\sqrt{6}} |0, 0\rangle + \frac{1}{\sqrt{6}} |1, 1\rangle$

$\therefore$ 基态: $\psi = \psi_1(r_1) \psi_1(r_2) \cdot \chi_3$ 六度简并

第一激发态: 九度简并

自旋反对称: $|1, 1\rangle = \frac{1}{\sqrt{2}} |1, 0\rangle - \frac{1}{\sqrt{2}} |0, 1\rangle$

$|1, -1\rangle = \frac{1}{\sqrt{2}} |0, -1\rangle - \frac{1}{\sqrt{2}} |-1, 0\rangle$

$|1, 0\rangle = \frac{1}{\sqrt{2}} |1, -1\rangle - \frac{1}{\sqrt{2}} |1, 1\rangle$

六、

(1) $H_0 = -\frac{1}{2} [(s_1 + s_2 + s_3)^2 - s_1^2 - s_2^2 - s_3^2] = -\frac{1}{2} [s(s+1) \hbar^2 - \frac{3}{4} \hbar^2 \times 3] = -\frac{1}{2} s(s+1) \hbar^2$

$\because \frac{1}{2} \otimes \frac{1}{2} = 0, 1 \quad 0 \otimes \frac{1}{2} = \frac{1}{2} \quad 1 \otimes \frac{1}{2} = \frac{1}{2}, \frac{3}{2} \quad \therefore S=0, \frac{1}{2}, 1, \frac{3}{2}$

$\therefore E_1 = E_2 = \frac{3\hbar^2}{4} \quad E_3 = -\frac{3\hbar^2}{4}$

(2)